MIL-P-60548A(MU)
AMENDMENT 1
10 September 1974

MILITARY SPECIFICATION
PROJECTILE, 105MM, SMOKE, WP, M60
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-60548A(MU), dated 22 May 1974, and is approved for use by all Departments and Agencies of the Department of Defense.

a. Paragraph 2.1, under Specifications add,

"MIL-M-6867 - Magnetic Inspection Units
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-STD-410- Certification of Magnetic Particle Inspection Personnel".

b. Add Paragraph 4.3.1.8:

"4.3.1.8 Magnetic Particle Inspection.-Heat treated or stress relieved bodies after final machining and prior to banding shall be inspected by the magnetic particle inspection technique approved by the responsible technical agency.

(a) 100 percent magnetic particle inspection will be in effect until 2500 consecutive items have been inspected and found to be free of metal defects.

(b) Reduced sampling may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

(c) If no defects are found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive days.

(d) If no defects are encountered during the five days, the sample may be reduced to five bodies per hour.

(e) If a defect is encountered by magnetic particle inspection during reduced sampling, (b through d), the magnetic particle inspection shall revert to (a) above, in addition, a 100 percent backscreening to the last acceptable hours production will be performed.

(f) For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

(g) Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible technical agency.

FSC 1315

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(h) Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternate procedure in lieu of corrective manufacturing process action to eliminate a continuing process induced defect. If corrective action is not instituted and the cause eliminated, the government reserves the right to withdraw acceptance inspection until the contractor can prove his process is capable of producing suspect free projectiles."

c. Add Paragraph 4.3.1.9:

"4.3.1.9 Magnetic particle (Inspection method).-The procedures for performing this test utilizing the wet fluorescent method shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be throughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. Magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL-M-6867. The operator(s) of the equipment and inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

d. Under Notes add paragraph 6.7:

"6.7 Because of the generally poorer surface quality of semi-finished steel purchased to ASTM A273 when compared to bar, the semi-finished materials should be selected for use with the hot cup-cold draw process only after careful review between steel supplier and projectile manufacture has indicated that the steel quality to be furnished by the steel supplier is compatible with the projectile manufacturers process and end item quality requirements. With regard to the manufacturing process, particular attention must be given to the amount of metal to be machined from the forged cup; since it is only with adequate metal removal at this processing operation that surface defects typically present in semi-furnished steel can be removed. If these surface defects are not machined off, problems can be anticipated in meeting the metal defects requirements of paragraph 3.6.1. "

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Preparing activity:

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Project No. 1315-A260

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